CRYSTAL
PRACTICE MANAGEMENT

CRYSTAL PM

HOW-TO GUIDE

Connect Crystal Meet rooms to Google Calendar

Give each room device read access to its own Google Workspace room calendar, so bookings show on the room page and the door sign and Join now joins them.

This guide has two halves. The Google half is done once for all rooms by someone with Google Workspace admin access and takes about 30 minutes. The device half takes about five minutes per room. Nothing here makes a room join a meeting by itself: someone in the room still presses Join now.

STEP 1
Create the rooms
Room resources in the Admin console.

STEP 2
Create the key
A service account and its JSON key.

STEP 3
Share the calendars
Each room calendar with the account.

STEP 4
Put it on the device
The key, the address, the check.

STEP 5
Book a test meeting
See it on the room page and sign.

BEFORE YOU BEGIN

Access. A Google Workspace super admin, or someone who can manage Buildings and resources and create a Google Cloud project.

Devices. The room devices installed as in **Set up a Crystal Meet room**, and their room names.

A way to copy one file to each device, such as `scp` from your computer or a USB stick.

The key is a password for the room calendars. Keep it out of email and chat, copy it straight to the devices, and delete the downloaded copy afterwards.

1 Create the rooms in Google Workspace

- **Open the Admin console** at `admin.google.com`, then **Directory › Buildings and resources › Manage resources**.
- **Add the building once.** Open **Buildings**, press **Add building**, fill in the name, address and floors, and save.
- **Add each room.** Open **Resources**, press **Add new resource**, choose category **Meeting space (room)** and type **Conference room**, pick the building and floor, and set the resource name to the same name the device uses, for example `Room 1`. Save, then repeat for Room 2 and Room 3.
- **Copy each room's calendar address.** Open the room and copy its **Resource email**. It looks like `c_1885a3b9c2d4e6f7g8h9i0j1k2l3m4n5@resource.calendar.google.com`. Keep the three addresses; step 4 needs them.

You should now see the three rooms under Resources, each with a resource email.

> **GOOD TO KNOW**
>
> Staff book a room by adding it to an event under **Rooms**. New rooms can take up to 24 hours to appear in that list, but the room's calendar exists straight away, so you can continue.

2 Create the service account and its key

- **Open the Cloud console** at `console.cloud.google.com`, open the project picker at the top, press **New project**, name it `Crystal Meet` and create it. Make sure it is selected.
- **Turn on the Calendar API.** Go to **APIs & Services › Library**, search for **Google Calendar API**, open it and press **Enable**.
- **Create the account.** Go to **IAM & Admin › Service accounts**, press **Create service account**, name it `crystal-meet-rooms`, press **Create and continue**, skip the roles, and press **Done**.
- **Create the key.** Open the account, go to **Keys**, press **Add key › Create new key**, choose **JSON** and create it. A file downloads; rename it `google-service-account.json`.
- **If Google says key creation is not allowed.** Newer Google Cloud organizations block service account keys by default. Go to **IAM & Admin › Organization policies** (you need the Organization Policy Administrator role, which a super admin can grant to their own account under IAM), open **Disable service account key creation**, press **Manage policy**, choose **Override parent's policy**, add a rule with enforcement **Off**, set the policy, then create the key again.
- **Copy the account's email.** It ends in `.iam.gserviceaccount.com`. Step 3 needs it.

You should now see the service account listed, and the key file on your computer.

3 Share each room calendar with the service account

- **Allow full sharing to outside addresses, once.** In the Admin console go to **Apps › Google Workspace › Calendar › Sharing settings**. Under **External sharing options for secondary calendars** choose **Share all information, and outsiders can change calendars** and save. Room calendars count as secondary calendars, and the service account is an outside address: the rooms book themselves from the door sign and the room page, so the account has to be able to add events. Nothing changes for anyone a calendar is not shared with.
- **Subscribe to the rooms.** Open `calendar.google.com`. Next to **Other calendars** press **+**, choose **Browse resources** and tick the three rooms. They appear under Other calendars.
- **Share each room.** Hover the room's name, open its menu, choose **Settings and sharing**, then under **Share with specific people or groups** press **Add people and groups**, paste the service account email, set the permission to **Make changes to events**, and send. Repeat for each room. A room shared with **See all event details** instead shows its bookings but cannot be booked at its screens: what is booked there is kept on the device until the sharing is changed.

  You should now see the service account listed under each room's sharing settings with Make changes to events.

4 Put the key and the calendar address on each device

- **Copy the key to the device.** From your computer, type `scp google-service-account.json pi@crystal-meet-room-1.local:~/` (use the device's name or IP address and your username), or carry it over on a USB stick.
- **A device you are installing now.** Before running the installer, open `~/room.yaml` and put the room's calendar address in `google_calendar_id`, keeping the quotes. Then run `sudo bash installer/install.sh --config ~/room.yaml --credentials ~/google-service-account.json` and start the service as the setup guide says.
- **A device that is already installed.** Type `sudo install -o pi -g pi -m 600 ~/google-service-account.json /etc/croom/google-service-account.json` (your username instead of pi), then `sudo nano /etc/croom/config.yaml`, set `google_calendar_id` to the room's address, save, and type `sudo systemctl restart croom`.
- **Check it.** Type `/opt/croom/venv/bin/croom --check-calendar -c /etc/croom/config.yaml`. It prints who it connected as, the room's calendar, and the next bookings:

```
Google Calendar: connected as crystal-meet-rooms@crystal-meet.iam.gserviceaccount.com
Calendar: Room 1 (c_1885a3b9c2d4e6f7g8h9i0j1k2l3m4n5@resource.calendar.google.com)
Bookings in the next 7 days:
  Thu Sep 25  9:00 AM to 9:30 AM   Design review          Zoom link
  Thu Sep 25  1:00 PM to 2:00 PM   Board lunch           no video link
```

- **Delete the copy in your home folder.** Type `rm ~/google-service-account.json`. The installed copy under `/etc/croom` is the one the room uses.

You should now see the check end with the room's name and its bookings, or "No bookings in the next 7 days", and the room page's schedule fills within a minute.

5 Book a test meeting

- **Create the booking.** In Google Calendar, make an event a few minutes from now, add the room under **Rooms**, and add a video link: a Google Meet, the Zoom add-on, or a pasted Zoom invitation link. A Zoom meeting's passcode can be in the link, in the invitation text pasted into the event ("Passcode: 123456"), or typed in the **Passcode** field the room page shows when Zoom asks for it. Save it.
- **Watch the room.** Within a minute the room page lists it under **Today** and the door sign shows it as next. Ten minutes before the start the door sign's bar and label turn amber (the photograph stays as it is) and the room page shows **Join now**.
- **Join and leave.** Press **Join now**; the TV joins the link. Press **Leave**, then **Tap again to leave**.

You should now see the booking on the room page and the sign, and the TV join when Join now is pressed.

> **TIP**
>
> If the room declines a booking (the event shows the room as declined), the slot was already taken by another booking. The device ignores declined bookings, so the sign stays right.

IF SOMETHING IS OFF

Troubleshooting

The check says "not configured"

It names the cause: no key path, a missing key file, or an empty or placeholder address. Check that `/etc/croom/google-service-account.json` exists, fix `/etc/croom/config.yaml`, then restart the service and run the check again.

The check says "not found or not shared"

The address is wrong or step 3 was missed for this room. Copy the resource email again and share the room calendar with the exact service account email. If Google refuses "Make changes to events", do the sharing setting at the start of step 3 first.

The check says "for reading only"

The room is shared with See all event details. Change the service account's permission on that room to **Make changes to events**. Bookings made at the room's screens in the meantime are sent on the device's next poll, within a minute.

The check says it could not sign in

The file is not a service account JSON key (download a new one from the account's Keys page), the Calendar API is not enabled in that project, or the key was deleted in the Cloud console.

A booking is missing on the room page

The room was not added to the event, the room declined it, or it is more than 7 days away. Cancelled bookings never show. Check the device's clock and time zone if today's bookings look shifted.

The booking shows but Join now is missing

The event has no video link the device can join. Add a Google Meet, use the Zoom add-on, or paste the Zoom invitation link. Its passcode can be in the link, in the invitation text, or typed in the room page's **Passcode** field.

Google says key creation is not allowed

The organization policy "Disable service account key creation" is on, which is the default for newer Google Cloud organizations. Turn it off for the Crystal Meet project as described in step 2, then create the key again.

Bookings show at the wrong time

Set the device's time zone in **Preferences › Raspberry Pi Configuration › Localisation**, then type `sudo systemctl restart croom`.

GOOD TO KNOW

The key can only read and book the calendars you shared with it, and it only ever adds, shortens or deletes the bookings made at the rooms' own screens. To take that away, delete the key on the account's Keys page. To replace it, create a new key and repeat step 4 on each device.